

Genetic Code & Translation - Factoid Sheet

Week 4 - Wednesday - Core Lecture 8 - current faculty deck + official Week 4 objectives

Master chain: code → charge tRNA → initiate at AUG → A/P/E elongation → stop/release → fold/modify/target protein.

Genetic Code

AUG = start/Met. Stop = **UAA, UAG, UGA**. mRNA read 5' → 3'; protein grows N → C. Code is degenerate but unambiguous.

Mutation Effects

Type	Result
Silent	Codon change, same amino acid
Missense	Different amino acid
Nonsense	Premature stop
Frameshift	Indel not divisible by 3; downstream frame changes
In-frame indel	3n bases; frame retained

tRNA

Aminoacyl-tRNA synthetase uses ATP to attach the correct AA to its tRNA. Wobble at codon position 3 explains why fewer tRNAs can decode many synonymous codons.

Initiation

Bacteria	Eukaryotes
Shine-Dalgarno + 16S rRNA	5' cap scanning + Kozak
fMet initiator	Met initiator
70S	80S

A/P/E

A accepts aminoacyl-tRNA; **P** holds peptidyl-tRNA; **E** = exit. Peptidyl transferase is rRNA-based.

Translation Regulation - Fast Recall

Global Initiation Control

eIF2 α -P traps limited eIF2B → GDP cannot recycle to GTP → initiation falls.

4E-BP binds eIF4E and blocks cap-dependent initiation; phosphorylation of 4E-BP releases eIF4E.

Iron Response

Iron	5' IRE: Ferritin / ALA synthase	3' IRE: Transferrin receptor
Low	IRP binds → translation OFF	IRP binds → mRNA stabilized
High	IRP releases → translation ON	IRP releases → mRNA degraded more readily

Viruses

IRES = internal ribosome entry site; permits cap-independent recruitment of host translation machinery.

Protein Destination

Free ribosome: cytosolic proteins + many post-translational imports.

RER-bound ribosome: secreted, lysosomal, and membrane proteins.

Post-Translational

Chaperones help folding; proteins may undergo cleavage, glycosylation, phosphorylation, acetylation, and signal-sequence-directed trafficking.

Must-Not-Miss

Shine-Dalgarno = prok. Kozak = euk. eIF2 phosphorylation shuts initiation down. Low iron = less ferritin, more transferrin receptor. IRES = cap-independent viral translation.

Assigned ScholarRx

DNA Transcription & Translation: <https://exchange.scholarrx.com/brick/dna-transcription-and-translation>